

Research Choices of Agentic Artificial Intelligence^{*}

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Abstract

Researchers increasingly rely on agentic artificial intelligence to assist with research. I examine the sampling and specification choices AI agents make and compare them to those of human experts. Fixing the research question and raw data across 156 agent runs, AIs tend to cluster on certain options. Compared to 145 human expert analyses of the same question and data, estimation outcomes are broadly similar. AI achieves this, however, via systematically different choices. Agentic AI thus remains an imperfect substitute for human researchers for the models tested.

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1 Introduction

Recent advances in artificial intelligence are transforming the way research is done. Economists increasingly use agentic AI, such as Claude Code and Codex, to clean and analyze data.¹ Some are testing the extent to which agentic AI can automate the entire research pipeline from idea to paper (Project APE, 2026; Willner and Yanagizawa-Drott, 2026). The rise of AI agent use makes verification of AI output the bottleneck: ensuring AI research choices are appropriate and match with the preferences of the human expert (Korinek, 2025). Systematic evidence on the research choices of AI, however, is lacking.

In this paper, I examine AI agents’ sampling and specification decisions and the corresponding estimation results. I then assess how well these choices compare to those of human experts. To this end, I direct 156 AI agents to answer the same question studied by many human analysts in Huntington-Klein *et al.* (2025): how eligibility for the Deferred Action for Childhood Arrivals program affects the probability of full-time employment for eligible workers. Each agent receives the same research prompt and raw dataset but independently generates a complete empirical specification—sample restrictions, estimating equation, and standard errors. The inherent stochasticity of the large language models that power modern AI agents ensures there is variation in research choices, though to what degree is not well understood.

Sampling and specification choices differ between AI agents and human experts. At many forking paths, such as which regression method to use or whether to use sampling weights, AI agents overwhelmingly make the same decision, despite AI agents’ underlying stochasticity. By contrast, there is more variation in the decisions human experts make at the same forking paths. Despite these differences, the AI agents’ distributions of estimation outcomes broadly resemble those of the human experts. Overall, my findings underline why researchers should review the data cleaning and analysis decisions of AI, as it is likely that some of these choices may not line up with one’s own preference.

A growing body of evidence shows that researchers answering the same question and using the same raw data make different decisions that can produce meaningfully different estimates

¹A straw poll during the NBER Summer Institute for Household Finance revealed approximately 60% of economists are ‘all-in’ on agentic AI, see https://x.com/S_Stantcheva/status/2080706102044291124.

(Silberzahn *et al.*, 2018; Huntington-Klein *et al.*, 2021; Breznau *et al.*, 2022). The present study is one of several this year to examine variation in agentic AI estimates: Gao and Xiao (2026), Grundl (2026), and Huang *et al.* (2026). Among these papers, only Grundl (2026) looks at the study by Huntington-Klein *et al.* (2025), and he does not focus on agent choices as I do.

2 Method

2.1 AI agents as empirical researchers

Large language models give non-deterministic responses: the same prompt can generate a range of responses. Submitting the same prompt to estimate a given empirical parameter many times therefore yields a distribution of AI choices.

AI agents may produce reasonable research choices because their underlying models are trained on huge corpora of digital data (OpenAI, 2025; Anthropic, 2026), almost surely including much economics research and associated code. The language in the prompt given to the AI agent (e.g., “causal” and a focus on labor market outcomes) should nudge output toward the conventions of applied economists, rather than, say, data science. AI outputs may therefore be interpreted as draws from a noisy representation of standard practices in economics. Alternatively, AI agents may make choices far outside disciplinary norms, or less internally coherent ones. Given model complexity, stochasticity, and closed-source nature, how closely AI agents hew to human patterns is an empirical question.

2.2 Application

I task AI agents with the same question answered by many human analysts in Huntington-Klein *et al.* (2025). The authors retained 146 teams of economists and asked each to independently answer the same causal question using the same instructions and raw data: how did eligibility for the Deferred Action for Childhood Arrivals immigration policy affect the probability of full-time employment among eligible immigrants? Participants were given a data codebook and the research question but otherwise initially exercised broad discretion

over sample construction, variable definitions, and specification choices (called Task 1 in the study).

The computing pipeline works as follows. An AI agent is provided with the raw materials to answer the research question: an IPUMS extract and codebook with a broad set of variables, a supplementary state policy dataset and codebook, and instructions adapted from [Huntington-Klein *et al.* \(2025\)](#).² The agent then writes a concise, structured file outlining their model specification, outcome definition, sample selection, and treatment definition. They proceed to write and execute a Python script to implement the estimation, working through any coding errors along the way, and reporting the resulting coefficient estimate, standard error, and sample size. Sampling and specification choices are then programmatically scraped from agents’ reports and cross-referenced against the actual code they produced to ensure accuracy.

Access to IPUMS differed between the two groups. Humans were instructed to navigate to the IPUMS website and select the variables they thought relevant. In contrast, AI agents received a fixed IPUMS extract containing all variables used by any of the human researchers, obtained by scanning the repository of human submissions to [Huntington-Klein *et al.* \(2025\)](#) hosted by Open Science Framework. This restricts AI agents from choosing any variable whatsoever, but still leaves substantial latitude in variable selection.³

I used two then-frontier medium-sized AI models—GPT-5.4 mini (OpenAI) and Claude Sonnet 4.6 (Anthropic)⁴—accessed through the GitHub Copilot CLI. I primarily rely on GPT-5.4 mini, as it is cheaper to run than Claude Sonnet 4.6. When using a subscription such as Codex or Claude Code, the marginal cost of each run is modest, making the exercise more affordable for individual researchers.

²Replication code and the AI agent results can be found at <https://github.com/brettmcc/LLM-bootstrapping/>.

³An alternative would have been to give each AI agent access to the IPUMS API via my personal API key. I opted against this both to avoid hundreds of large ACS files on my machine and to not overburden the IPUMS API.

⁴I exclude 40 exploratory runs with Claude’s smaller and less advanced Haiku 4.5 model, which yielded several degenerate results, such as a standard error of 0. Researchers applying this approach should therefore take heed that weaker models may not be suitable for this type of task.

3 Results

I explore 156 AI agent analyses and compare them to 145 human analyses from [Huntington-Klein *et al.* \(2025\)](#). Both humans and agents had substantial freedom to over sample selection, variable definitions, and specification.

3.1 Choices of AI and Humans

Table 1 displays the prevalence of model specification choices between AI and human runs.

Table 1: Estimation choices by researcher type

Category	Choice	AI		Humans	
		N	Share (%)	N	Share (%)
Method	Linear Regression	156	100.0	122	84.1
	Logit/Probit	0	0.0	19	13.1
	Matching	0	0.0	2	1.4
	New DID Estimator	0	0.0	1	0.7
	Other	0	0.0	1	0.7
S.E. Adjustment	Cluster (State)	127	81.4	31	21.2
	Cluster (State & Year)	1	0.6	22	15.1
	Cluster (ID/Strata/Other)	0	0.0	20	13.7
	Het-Robust	26	16.7	24	16.4
	Other/Bootstrap	0	0.0	7	4.8
	None	2	1.3	42	28.8
Weights	No Sample Weights	7	4.5	101	69.2
	Sample Weights	149	95.5	45	30.8

Notes: Estimation choices are inferred from each generated model specification and execution metadata. Human results use Task 1 submissions from [Huntington-Klein *et al.* \(2025\)](#).

I find that AI agents exhibit less variation in specification choices than humans. AI agents exclusively chose linear models, in contrast to 13% of human runs using nonlinear models. AI agents also overwhelmingly converged on clustering standard errors at the state level, with 81% of agents choosing to do so compared to 21% of human runs. The share of runs featuring heteroskedasticity robust standard errors was quite similar between AI and humans.

One concern is that differences the programming language used may contribute to differences

in research choices. Most humans used Stata or R, while I required the AI to use Python. Default standard error computation differs across languages, and AI may go with default settings. Instead, 98.7% of AI agents actively chose a standard error method rather than relying on the default.⁵ What I cannot rule out is that observed empirical practices may have differed by language in the AI model’s training data, and these differences may affect the observed choices.

AI agents almost exclusively used sample weights, while a majority of human researchers did not. Using weights is in line with IPUMS instructions, a fact highlighted by [Huntington-Klein *et al.* \(2025\)](#).

Choice of control variables differed substantially between humans and AI, as shown in Table 2. AI agents included an age control more frequently than humans did, and AI almost never controlled for education while humans did in nearly half of specifications. 92% of AI agents included year fixed effects, compared to just 27% of human runs. Similarly, 92% of AIs included state fixed effects compared to 36% of human runs. Estimating a fully-saturated difference-in-differences specification diminishes the downsides of linear probability models, the choice of every AI agent, as noted by [Huntington-Klein *et al.* \(2025\)](#). Overall, AI agents agreed on the set of controls much more often than human researchers. Just 25% of AI agent runs used a set of controls unique among all other AIs, compared to 60% of humans.

⁵This contrasts with the excluded AI runs executed by the less intelligent Claude Haiku 4.5, which heavily relied on the default standard error method.

Table 2: Control variable choices by researcher type

Category	Control	AI		Humans	
		N	Share (%)	N	Share (%)
AGE	Linear Age	82	52.6	61	41.8
AGE	Age FE	19	12.2	13	8.9
AGE	Age Quadratic	69	44.2	11	7.5
EDUC	Linear Education	3	1.9	46	31.5
EDUC	Education FE	1	0.6	8	5.5
EDUC	Education Transform	0	0.0	16	11.0
STATE/YEAR	Linear Year	16	10.3	33	22.6
STATE/YEAR	Year FE	143	91.7	40	27.4
STATE/YEAR	State FE	144	92.3	52	35.6
STATE/YEAR	State FE x Year FE	132	84.6	19	13.0
STATE/YEAR	State FE x Linear Year	15	9.6	5	3.4

Notes: Each block reports how runs restricted the variable named in its heading; categories within a block are mutually exclusive and sum to the block total. "All" columns report restrictions on the estimation sample, "Treated" columns the restrictions used to code DACA eligibility, which inherit those used to restrict the whole sample. YRSUSA is the IPUMS years-in-the-US variable. The table compares AI-agent restriction choices with the published Task 1 counts from Table 6 of [Huntington-Klein *et al.* \(2025\)](#).

Next, Table 3 compares sample and treated group filters. Rows underneath each variable heading are roughly ordered by the ‘correctness’ for defining the treated group, as in [Huntington-Klein *et al.* \(2025\)](#). This is based on DACA’s eligibility criteria, which are listed in the AI’s prompt. The prompt does not specify, however, how to construct the control group.⁶

Table 3: Sample and treated-group restriction choices by researcher type

Variable	AI				Humans			
	All		Treated		All		Treated	
	N	Share (%)	N	Share (%)	N	Share (%)	N	Share (%)
Hispanic	156		156		144		144	
... Hispanic-Mexican	154	98.7	154	98.7	105	72.9	109	75.7
... Hispanic-Any	2	1.3	2	1.3	17	11.8	17	11.8
... Hispanic-Mex or Mex-Born	0	0.0	0	0.0	1	0.7	2	1.4
... None	0	0.0	0	0.0	21	14.6	16	11.1
Birthplace	156		156		145		145	
... Mexican-Born	154	98.7	154	98.7	103	71.0	112	77.2
... Hispanic-Mex or Mex-Born	0	0.0	0	0.0	2	1.4	2	1.4
... Non-US Born	0	0.0	0	0.0	4	2.8	4	2.8
... Central America-Born	0	0.0	0	0.0	1	0.7	1	0.7
... None	2	1.3	2	1.3	35	24.1	26	17.9
Citizenship	156		156		145		145	
... Non-Citizen	153	98.1	153	98.1	83	57.2	117	80.7
... Foreign-Born	2	1.3	2	1.3	2	1.4	2	1.4
... Non-Cit or Natlzd post-2012	0	0.0	0	0.0	4	2.8	7	4.8
... Other	1	0.6	1	0.6	11	7.6	11	7.6
... None	0	0.0	0	0.0	45	31.0	8	5.5
Age at Migration	156		156		145		145	
... < 16	56	35.9	112	71.8	21	14.5	105	72.4
... ≤ 16	0	0.0	0	0.0	10	6.9	25	17.2
... Other	36	23.1	12	7.7	24	16.6	11	7.6
... None	64	41.0	32	20.5	90	62.1	4	2.8
Age in June 2012	156		156		145		145	
... Year-Quarter Age	6	3.8	11	7.1	40	27.6	117	80.7
... Year-Only Age	148	94.9	143	91.7	18	12.4	21	14.5
... Other	0	0.0	0	0.0	2	1.4	0	0.0
... None	2	1.3	2	1.3	85	58.6	7	4.8
Year of Immigration	156		156		145		145	
... < 2007	0	0.0	0	0.0	15	10.3	43	29.7
... ≤ 2007	78	50.0	133	85.3	13	9.0	52	35.9
... < 2012	0	0.0	0	0.0	3	2.1	1	0.7
... ≤ 2012	0	0.0	0	0.0	2	1.4	4	2.8
... Any Year	44	28.2	9	5.8	7	4.8	4	2.8
... Other	26	16.7	9	5.8	5	3.4	3	2.1
... None	8	5.1	5	3.2	100	69.0	38	26.2
Years Continuous in USA	156		156		145		145	
... Used YRSUSA	1	0.6	4	2.6	23	15.9	55	37.9
... No YRSUSA	155	99.4	152	97.4	122	84.1	90	62.1

Notes: The table compares AI-agent restriction choices with the published Task 1 counts from Table 6 of [Huntington-Klein *et al.* \(2025\)](#). Treated-group restrictions inherit those used to restrict the whole sample.

⁶In prompting the AI, I did not mention restricting the sample based on education or veteran status, consistent with the prompt given to the human researchers in Task 1 by [Huntington-Klein *et al.* \(2025\)](#).

Once again AI agents overwhelmingly converged on several sample and treated-group construction choices. Virtually all AI agents restricted the sample to ethnically Hispanic-Mexicans born in Mexico who are not US citizens.⁷

Moreover, AI consistently diverged from humans when coding treatment. First, only 7% of AI runs coded age from both birth year and birth quarter, against 81% of human runs, relying instead on age in years alone. Yet birth year alone cannot say whether someone born in 1981 had turned 31 by June 15, 2012, the program’s age cutoff, so treatment eligibility is more often misassigned by AI. Second, DACA required continuous US residence since June 15, 2007, yet no AI run required arrival strictly before 2007 (versus humans’ 30%), and 85% instead admitted arrivals through the end of 2007 (versus humans’ 36%). Since the data record only the year of arrival, neither cutoff is exact, but the human choice is the more conservative one, excluding ineligible immigrants who may have arrived after June 15, 2007 rather than admitting them.⁸ Hence AI is less conservative on average than humans in assigning DACA eligibility, including more ineligible individuals in the treated group.

Which group made the more defensible research choices? Although many research choices cannot be objectively ranked by ‘correctness’, some are clearly more appropriate than others. IPUMS recommends using sample weights; adjusting standard errors for heteroskedasticity or cluster dependence is more conservative; national shocks are a common confounder addressed by year fixed effects; and the prompt explicitly targets the Hispanic Mexican-born population. On each of these dimensions, AI more consistently made the defensible choice. Yet AI underperformed on coding up age and year treatment cutoffs. In still other cases, AI’s choice was not ex-ante obviously better, such as all AIs estimating a linear regression. Thus, AI’s greater convergence partly reflects concentration on better-supported choices, but not uniformly so. A comprehensive assessment of the quality of AI research choices is beyond the

⁷I suspect this results from the AI becoming highly focused on the first clause of the AI prompt (copied verbatim from the human task), “**Among ethnically Hispanic-Mexican Mexican-born people living in the United States**, what was the causal impact of eligibility for the Deferred Action for Childhood Arrivals (DACA) program (treatment) on the probability that the eligible person is employed full-time (outcome),” emphasis mine.

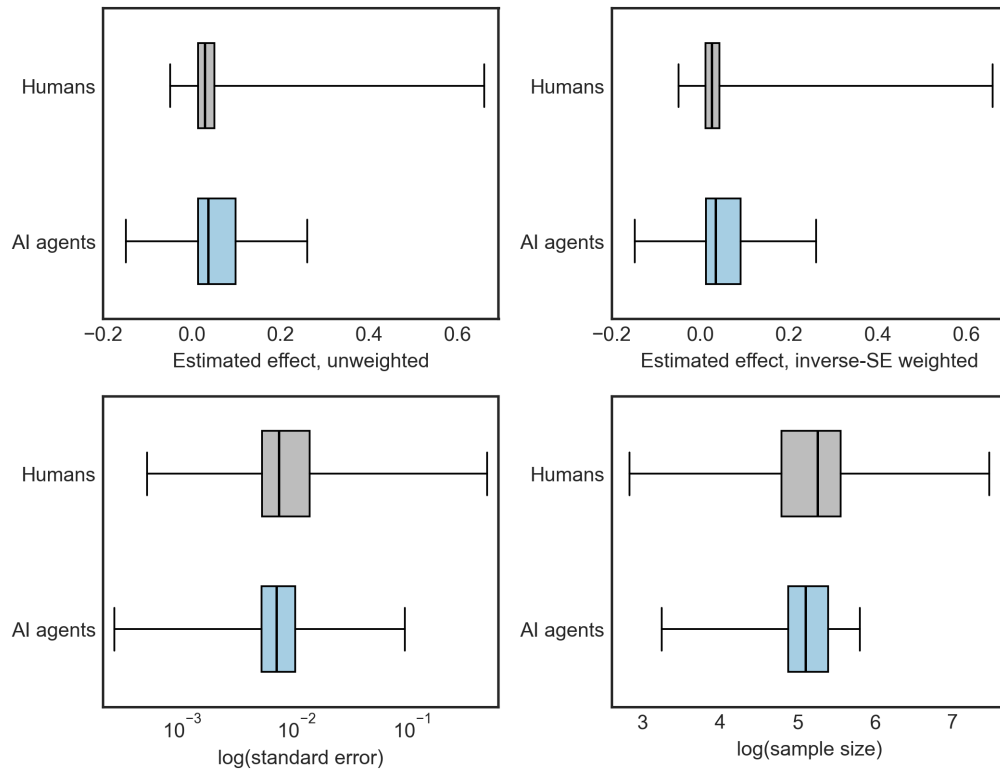
⁸AI rarely used years of continuous US residence in defining treatment (3% vs. humans’ 38%). The IPUMS variable YRSIMMIG simply equals the difference between the survey year and the reported year of immigration, however, so it is perfectly substitutable with Year of Immigration.

scope of this paper and remains an important topic for future work.

3.2 Distribution of estimates

Despite sharp differences in sampling and specification choices, how do the resulting estimation outcomes compare? Figure 1 depicts box-and-whisker plots for both human and AI analyses for weighted and unweighted effects, standard errors, and sample size. Exact values are reported in Table 4.

Figure 1: Human and AI estimation outcomes



Notes: Panels compare unweighted point-estimate distributions (top left), point-estimate distributions weighted by inverse standard error (top right), standard errors (bottom left), and log sample sizes (bottom right) for both the human researcher sample of Task 1 from [Huntington-Klein et al. \(2025\)](#) and the AI agent sample from this paper. In line with [Huntington-Klein et al. \(2025\)](#), inverse standard error weights are capped at the 95th percentile. Boxes show the interquartile range, center lines show medians, and whiskers show minimum and maximum values.

Across all four outcomes, medians are fairly similar between human and AI researchers. Moreover, AI agents exhibit substantial dispersion in their estimation outcomes, underlining the stochasticity of AI outputs.

For the unweighted effect size plotted in the top left, AI runs yield a mean of 0.064 and

Table 4: Summary statistics on estimation outcomes

	N	Mean	SD	Min	Pctl. 25	Median	Pctl. 75	Max
<i>Panel A: AI-agent estimates</i>								
Effect Size (unweighted)	156	0.064	0.084	-0.148	0.014	0.037	0.099	0.261
Effect Size (weighted by inverse SE)	156	0.060	0.080	-0.148	0.013	0.035	0.091	0.261
Standard Error	156	0.008	0.007	0.000	0.005	0.007	0.010	0.088
Sample Size	156	165,965	123,732	1,764	75,319	125,655	248,270	636,722
Treated-Group Size	155	65,518	28,865	7,717	36,477	74,231	81,508	169,320
<i>Panel B: Human estimates</i>								
Effect Size (unweighted)	145	0.053	0.095	-0.049	0.014	0.030	0.051	0.660
Effect Size (weighted by inverse SE)	138	0.044	0.092	-0.049	0.012	0.026	0.043	0.660
Standard Error	139	0.019	0.055	0.000	0.005	0.007	0.013	0.460
Sample Size	145	828,318	3,056,037	681	61,600	179,960	356,787	29,536,580
Treated-Group Size	141	96,395	648,493	270	17,950	34,435	52,581	7,727,201

Notes: Panel A reports outcomes from the AI agent sample. Panel B restates the published Task 1 panel of Table 3 of [Huntington-Klein *et al.* \(2025\)](#). Treated-group sample size not available for all AI agent runs since its collection was added after the initial runs.

a median of 0.037 and humans a mean of 0.053 and median of 0.030. The 25th percentile estimates are identical at 0.014, while the 75th percentile effect is substantially larger for AI agents at 0.099 relative to humans’ 0.051. Results are quite similar for the inverse-standard error weighted coefficients plotted in the top right.

Human point estimates exhibit a 13% higher standard deviation in effect size, driven by a much longer positive tail. The maximum human estimate is 2.5 times the maximum AI estimate, while the minimum AI estimate is 3 times more negative than the minimum human estimate. Still, 92% of AI point estimates are within the min-max range of humans, and 30% are within humans’ interquartile range.

For standard errors, plotted in the bottom left, the minimum, 25th percentile, and median are identical across humans and AI, while human estimates exhibit higher standard errors at the 75th percentile and maximum. Correspondingly, AI agents obtain coefficients statistically significantly different from 0 about 90% of the time, compared to 78% of the time among humans.

The range of sample sizes chosen by human researchers—between 681 and over 29 million observations—is several orders of magnitude larger than what the AI agents chose. Still, the interquartile range is comparable: 75 to 248 thousand for the AI, and 61 to 357 thousand for

the humans.

4 Conclusion

This paper assesses the choices of agentic artificial intelligence and how they compare to human experts'. I do so leveraging the multi-analyst study of [Huntington-Klein *et al.* \(2025\)](#).

While AI agent estimation outcome distributions resemble those of humans, the way in which each group arrives at those estimates differ markedly. AI agents converged much more on particular specification and sampling choices relative to human researchers. This finding is consistent with concurrent work by [Huang *et al.* \(2026\)](#), who find in a different setting that AI tends to cluster on certain sampling and specification choices. Hence, I conclude that AI is thus far an imperfect substitute for human experts.

Two limitations are worth emphasizing. First, the [Huntington-Klein *et al.* \(2025\)](#) materials were public before my runs,⁹ so model training data may have included the original study or analyst submissions. [Karmakar *et al.* \(2022\)](#), [Zhang *et al.* \(2023\)](#), and [Al-Kaswan *et al.* \(2026\)](#) show that exposure to materials during training may shape model responses without necessarily regurgitating training data verbatim. Still, the convergence of many AI choices in contrast to the human analysts indicate the AI was not merely copying human work. Second, results may depend on agent model and harness. Future work should examine whether more advanced models, such as (at the time of writing) GPT-6 and Claude Fable 5.1 can better match or even outperform human experts.

References

- AL-KASWAN, A., SPIESS, C., DEVANBU, P., VAN DEURSEN, A. and IZADI, M. (2026). Model see, model do? exposure-aware evaluation of bug-vs-fix preference in code llms. In *Proceedings of the 23rd International Conference on Mining Software Repositories*, pp. 86–97.

⁹See <https://osf.io/9p7j6/files/osfstorage>.

- ANTHROPIC (2026). System card: Claude sonnet 4.6.
- BREZNAU, N., RINKE, M. E., WUTTKE, A. *et al.* (2022). Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty. *Proceedings of the National Academy of Sciences*, **119** (44), e2203150119.
- GAO, R. and XIAO, S. C. (2026). Nonstandard errors in AI agents. SSRN working paper, date written: March 16, 2026.
- GRUNDL, S. (2026). A comparison of agentic AI systems and human economists. SSRN working paper, date written: April 9, 2026.
- HUANG, W., MENKVELD, A. J. and YU, S. (2026). AI “errors”. SSRN working paper, date written: March 13, 2026.
- HUNTINGTON-KLEIN, N., ARENAS, A., BEAM, E., BERTONI, M., BLOEM, J. R., BURLI, P., CHEN, N., GRIECO, P., EKPE, G., PUGATCH, T. *et al.* (2021). The influence of hidden researcher decisions in applied microeconomics. *Economic Inquiry*, **59** (3), 944–960.
- , PORTNER, C. C., MCCARTHY, I. and THE MANY ECONOMISTS COLLABORATIVE ON RESEARCHER VARIATION (2025). *The Sources of Researcher Variation in Economics*. I4R Discussion Paper 209, Institute for Replication (I4R).
- KARMAKAR, A., PRENNER, J. A., D’AMBROS, M. and ROBBES, R. (2022). Codex hacks hackerrank: Memorization issues and a framework for code synthesis evaluation. *arXiv preprint arXiv:2212.02684*.
- KORINEK, A. (2025). *AI agents for economic research*. Tech. rep., National Bureau of Economic Research.
- OPENAI (2025). GPT-5 system card.
- PROJECT APE (2026). Project APE: Autonomous policy evaluation. Social Catalyst Lab, University of Zurich, accessed July 29, 2026.
- SILBERZAHN, R., UHLMANN, E. L., MARTIN, D. P. *et al.* (2018). Many analysts, one

- data set: Making transparent how variations in analytic choices affect results. *Advances in Methods and Practices in Psychological Science*, **1** (3), 337–356.
- WILLNER, O. and YANAGIZAWA-DROTT, D. (2026). Verifying the verifiers: Towards autonomous policy evaluation, working paper, July 20, 2026.
- ZHANG, C., IPPOLITO, D., LEE, K., JAGIELSKI, M., TRAMÈR, F. and CARLINI, N. (2023). Counterfactual memorization in neural language models. *Advances in Neural Information Processing Systems*, **36**, 39321–39362.